

Intelligent Audio-Video Dubbing with Neural Lip Movement Synthesis Using Wav2Lip

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Abstract—Conventional dubbing techniques used in film, education, and broadcasting are often costly, time-consuming, and unable to achieve natural alignment between speech and lip movements. Recent advancements in deep learning have enabled intelligent systems that automate and enhance this process. This work proposes an audio-video dubbing framework that integrates Automatic Speech Recognition (ASR), Machine Translation (MT), Text-to-Speech (TTS), and neural lip movement synthesis using the Wav2Lip model. The system converts the source audio into text, translates it into the target language, and generates a corresponding speech waveform. To produce realistic and synchronized outputs, Wav2Lip aligns the synthesized audio with the lip movements of the speaker in the video. Experimental results demonstrate that the proposed method achieves high lip-sync accuracy, generates natural-sounding speech, and enhances accessibility across multiple Indian languages. This AI-driven approach reduces dubbing costs, improves multilingual viewing experiences, and bridges communication gaps across diverse language communities.

Index Terms—Audio-video dubbing, lip synchronization, Wav2Lip, speech synthesis, deep learning, machine translation.

I. INTRODUCTION

Audio Visual Translation (AVT) has come to the fore recently because of the proliferation of digital content consumption worldwide [1]. It enables access through the transfer of speech from one language to another without affecting the message conveyed and the consistency of visuals [2]. Dubbing is one such type that has been the most used translation technique since it offers the viewer an uninterrupted viewing experience. However, conventional methods of dubbing have proven to be very expensive, difficult, and time-consuming.

A. Audio-Visual Translation (AVT)

Audio-Visual Translation (AVT) involves translating the multimedia texts in order to make them accessible across languages through verbal transfer with the message conveyed and the accompanying visual context intact [2]. This field includes translation techniques like dubbing, subtitling, and voiceovers. Dubbing becomes the dominant mode within films and other forms of entertainment because of its ability to offer viewers an immersive experience. Dubbing allows for an easy blending of translated texts and is hence an important tool used in breaking down language barriers.

B. Challenges in Traditional Dubbing

The traditional method of dubbing is still an expensive process that involves intensive manual work, voice artists, and post-processing editing. These factors hamper scalability especially when it comes to multimedia projects on a large scale. Another factor is the use of English language for digital content since English makes up 60 percent of digital content but only a few percentage of people in the world have English as their native language. For example, in India where only about 10.67 percent of the people can understand English, there is a great need for localization considering the various languages used in the country.

C. Importance of Lip Synchronization

A critical factor determining the legitimacy of a dubbed film lies in the process of lip synchronization, which involves aligning spoken utterances with the corresponding mouth movement. Poor synchronization may cause disconnection between what the character is saying and their mouth movement, affecting the overall immersion experience. Traditional

techniques usually involve tedious manual work or are highly restrictive, hindering their applicability across various language pairs and speaking styles. Recently, advances in deep learning and computer vision have allowed for automatic lip synchronization with impressive results, creating convincing lip movement [4]. However, there is still a need for an integrated system that incorporates lip synchronization, ASR, MT, and TTS.

II. LITERATURE SURVEY

This section reviews relevant literature related to key aspects of the proposed system, including attention-based machine translation, Indian language speech processing, lip synchronization, automatic speech recognition, prosody modeling, audio-visual synchronization, and speech enhancement.

A. Attention Mechanism for Neural Machine Translation

The paper introduces the Transformer model that replaces recurrent models with an entirely attention-based encoder-decoder architecture. Specifically, the multi-head self-attention allows modeling of both short and long-range interactions simultaneously, resulting in faster convergence and improved translation accuracy compared to LSTM-based systems [1]. This literature is useful for the implementation of Transformer-based IndicTrans2 for Hindi-to-Marathi translation in our pipeline. However, the original Transformer model was designed for text-only applications and does not solve problems associated with prosodic alignment and audio-visual synchronization.

B. High-Quality Speech Translation for Indian Languages

The paper proposes IndicTrans2 – a multilingual neural machine translation framework trained on vast parallel data resources involving 22 Indian languages [2]. Due to its ability to accommodate agglutinative morphology, code-switching, and linguistic variation typical of Hindi, Marathi, Tamil, and Telugu, the model is particularly suited for the current task. According to evaluations on the FLORES benchmark, IndicTrans2 achieves state-of-the-art BLEU performance on low-resource Indian language pairs [2]. Further improvements could include the preservation of style during translation that will be partly achieved using the current voice cloning module[9][10].

C. Automated Lip Sync Detection in Unconstrained Video

The authors propose SyncNet – a two-stream CNN trained to learn a joint audio-visual embedding space with contrastive loss for assessing the consistency between visual lip movement and audio [3]. After training on the BBC lip-reading corpus, SyncNet demonstrates a high detection rate for lip-speech desynchronization cases. SyncNet serves as a discriminator in our pipeline for evaluation of generated videos' quality during inference. Notably, the method does not provide any tools for generating or repairing lip movement which is achieved through Wav2Lip.

D. Wav2Lip: Accurate Lip-Syncing with GAN

Wav2Lip is a GAN-based method developed by Prajwal et al.[4] that produces precise lip movements for any individual with an arbitrary audio track. The architecture includes two CNN-based encoders for facial identity and audio, followed by a decoder with audio-video alignment enforced by the SyncNet-based discriminator. Wav2Lip, trained on LRS2-BBC, has near-perfect lip synchronization, measured by LSE-D and LSE-C scores. Wav2Lip in our system takes MFCCs from the Marathi TTS audio file and mouth region frames from the source video. Known limitations of the model include its vulnerability to noise in the input audio, countered in our pipeline through noise suppression techniques [8].

E. Wav2Vec 2.0: Self-Supervised Speech Representation Learning

Wav2Vec 2.0 [5], an architecture that uses a CNN encoder followed by a Transformer context network trained via a contrastive learning approach on masked speech audio, was proposed by Baevski et al. When fine-tuned using only 10 minutes of labeled speech data, it shows good word error rates along with robustness against accent and rate differences. Wav2Vec 2.0, fine-tuned using Hindi language, is used in our ASR system as the main transcription model for noisy dialectal speech samples.

F. Prosody Modeling for Cross-Lingual TTS

Gudmalwar et al.[6] examined prosody transfer during cross-language synthesis of TTS, demonstrating that language switching naively will not help retain the inherent prosody of the source speech. The authors introduced a prosody encoder which was conditioned on the pitch and duration of the source speech, allowing more effective retention of the speaking style across languages. This study relates to our work since the TTS output in Marathi should reflect the emotions conveyed by the source Hindi speech. Prosody transfer in its full form, as discussed in [6], is our goal for the future.

G. Learning Audio-Visual Speech Synchronization

In their work [7], Zhou et al. suggest learning audio-visual correspondence in wild videos by means of two-stream convolutional networks based on contrastive learning. Contrary to SyncNet [3], this method is capable of handling arbitrary head positions, light variations, and multiple speakers. Synchronization score maps are employed in our system both for training Wav2Lip by means of an additional objective function and verification of lip-sync quality during inference.

H. RNNoise: Real-Time Full-Band Speech Enhancement

Ain [8] introduced RNNoise, a hybrid algorithm based on DSP and GRU that utilizes a recurrent neural network for the purpose of denoising real-time full-band speech signals. With a sampling rate of 48 kHz and a negligible amount of computational load, it filters out any unwanted background noises, reverberation effects, and recording distortions while maintaining the quality of speech.

TABLE I
COMPARATIVE SUMMARY OF RELATED WORKS

Ref.	Authors / Year	Technique	Task	Key Contribution	Limitation
[1]	Vaswani et al. (2017)	Transformer / Attention	MT & NLP	Introduced self-attention for sequence-to-sequence tasks; foundation for modern NMT	No speech or audio integration
[2]	Ramesh et al. (2022)	IndicTrans2 / NMT	Low-resource Indian language MT	High-quality Hindi-Marathi and regional language translation using multilingual NMT	Limited prosody and emotion handling
[3]	Chung & Zisserman (2016)	SyncNet / CNN	Lip sync detection	Automated audio-visual lip-sync detection in unconstrained in-the-wild videos	No dubbing or audio generation capability
[4]	Prajwal et al. (2020)	Wav2Lip / GAN	Lip synchronization	Accurate lip-sync generation for any speaker from arbitrary audio using GAN	Sensitive to noisy audio and large head movements
[5]	Baevski et al. (2020)	Wav2Vec 2.0 / Transformer	ASR	Self-supervised ASR robust to accents, variable speech rates, and moderate noise	Requires fine-tuning for low-resource languages
[6]	Gudmalwar et al. (2019)	Prosody Modeling / DNN	Cross-lingual TTS	Improved prosody and rhythm transfer in cross-lingual speech synthesis	Limited multi-speaker generalization
[7]	Zhou et al. (2021)	Two-stream CNN / Contrastive Loss	Audio-visual sync	Learns audio-visual synchronization from unconstrained video at scale	Discriminator only; cannot correct video desynchrony
[8]	Valin (2018)	RNNoise / DSP + GRU	Speech enhancement	Real-time full-band speech denoising with very low computational overhead	Degrades on very low SNR recordings

III. PROPOSED WORK AND TECHNOLOGIES USED

The proposed system provides end-to-end dubbing from Hindi to Marathi, maintaining speaker identities and achieving lip synchronization. The processing pipeline separates the audio and video parts, with ASR for text transcription, NMT for translation, and a speaker-dependent Text-to-Speech Synthesis for Marathi speaking. The audio quality can be improved using RNNoise, and lip synchronization can be done using Wav2Lip with SyncNet discriminator for generating realistic dubbing.

A. Automatic Speech Recognition (ASR)

The ASR system is responsible for converting Hindi speech into text in the first step of dubbing and contains two variants:

- 1) Web Speech API provided by Google provides Hindi cloud-based ASR with confidence scores that can be used in prototyping and clean audio operation.
- 2) Wav2Vec 2.0 [5] uses CNN encoder and Transformer architecture with Connectionist Temporal Classification (CTC), which can be fine-tuned on Indian language data to achieve robust transcription in noisy conditions and different dialects.

B. Translation Module

Translation module translates Hindi text into Marathi based on Neural Machine Translation (NMT) models [1]:

- 1) Google translate API was used at the beginning of development for rapid system integration and prototyping.
- 2) IndicTrans2/M2M-100 [2] are specifically developed for Indian languages. As a result, they provide accurate translations due to their ability to catch grammar, word order, and other linguistic properties of the language.

Length control is introduced to allow temporally accurate Marathi translations that are required for lip synchronization.

C. TTS (text-to-speech) systems

The TTS system synthesizes Marathi speech from translated text with the use of:

- 1) Neural TTS systems like Tacotron2, FastSpeech, FastSpeech2, Glow-TTS, VITS [11][12] generate high-quality mel-spectrograms.
- 2) HiFi-GAN [13] generates prosodically natural speech waveforms from the synthesized spectrograms.
- 3) Speaker embeddings obtained from the source Hindi audio [10] can condition cross-lingual voice characteristics.
- 4) There are lightweight offline models for Marathi TTS like gTTS and pyttsx3.

D. Voice Cloning Module

Voice cloning is responsible for keeping the original speaker's voice characteristics. Speaker vectors, containing timbre, pitch, speaking style, and other attributes, can be extracted from the source Hindi audio with the help of ECAPA-TDNN [9] and then condition the TTS model in order to synthesize Marathi speech with the same voice features.

E. Noise Reduction Module

The Noise Reduction module improves audio clarity and professional quality.

- 1) It uses RNNoise, which utilizes machine learning-based denoising algorithm that combines the power of recurrent neural network and signal processing.
- 2) RNNoise removes background noise and echoes without degrading the voice.
- 3) It is especially useful in real-life video scenarios like lectures, interviews, and outdoor videos.

F. Lip Synchronization Model (Wav2Lip)

For lip synchronization, Wav2Lip works by:

- 1) Transforming the Marathi audio into MFCC and performing facial region extraction frame-wise.
- 2) Using dual encoders based on CNN for processing audio and visual inputs before decoding to produce lip-synchronized frames.
- 3) Minimizing the objective function using L1 reconstruction loss and adversarial loss.
- 4) Learning using LRS2-BBC dataset by adapting it to Indian faces.

G. Lip Sync Discriminator (SyncNet-based)

Discrimination between well and poorly synchronized audio-visual pairs is carried out using SyncNet-based discriminator [7] which includes:

- 1) Dual-stream CNN where each stream processes lip motion and audio inputs independently.
- 2) Contrastive loss for distinguishing synchronized and unsynchronized audio-video pairs.
- 3) The obtained similarity score used both in training and testing.

IV. ARCHITECTURE

The architecture consists of inter-dependent sub-systems designed in such a way that they can work together in parallel pipelines, thereby providing support for processing audio and video components together in order to create automated lip synchronized dubbing videos.

A. Input Acquisition and A/V Decomposition

The process starts by acquiring raw video and then using FFmpeg for separating HD frames capturing facial expressions and mouth motion along with a clean speech signal ready for speech representation.

B. Audio Pre-processing and Speech Representation

Two kinds of pre-processing are used for generating text from audio. One involves reducing noise by using RNNoise or Spectral Subtraction [8]. The other step involves speech transcription via ASR using Whisper or Wav2Vec 2.0 [5] and obtaining transcribed text along with its timings.

C. Machine Translation and Semantic Conversion

After obtaining the transcription text, it is further translated in the target language using transformer based NMT algorithms such as IndicTrans and IndicTrans2 [1][2]. Transcription text with semantic meaning will be generated as part of the process required in further steps.

D. Speech Generation via TTS and Voice Cloning

For this process, two techniques will be applied together. Firstly, neural TTS technology like HiFi-GAN and VITS [13] will be applied for high quality speech generation. At the same time, speaker embedding [10] technique will be used for recreating the voice characteristics of the original speech.

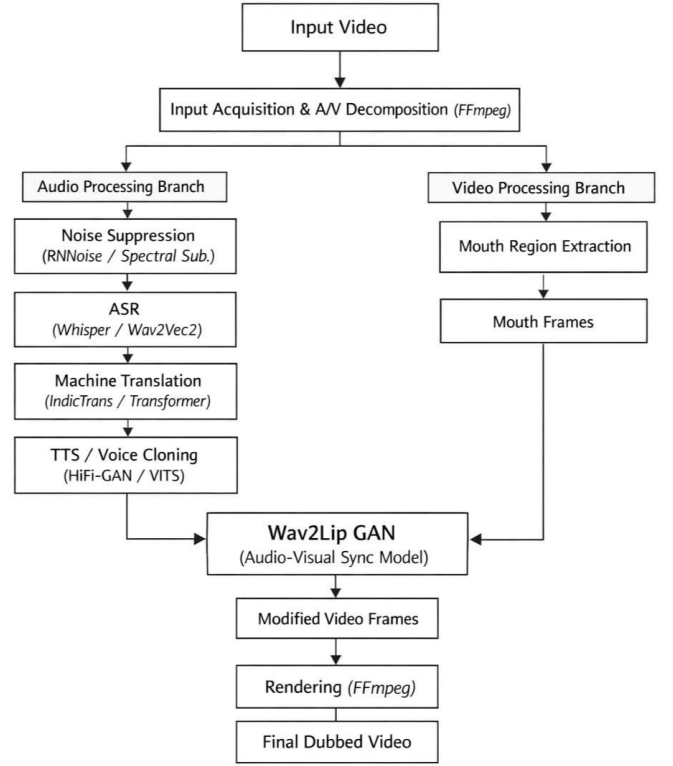


Fig. 1. Full System Architecture for Proposed Lip-Sync Dubbing System.

E. Visual Mouth-Region Processing

This process involves extracting video frames containing mouths using image-processing techniques. These mouth frames will be used as the visual component in the lip synchronization model.

F. Lip-Synchronization via Wav2Lip GAN

Wav2Lip[4] will merge the newly created Marathi audio with mouth frames to produce frames of video with lip synchronization through the use of GAN Architecture.

G. Rendering and Output Reconstruction

Final output video synthesis can be done using FFmpeg by combining both audio and video components.

V. RESULTS AND DISCUSSIONS

In this section, both quantitative and qualitative evaluations of the AI-Based Multilingual Dubbing and Neural Lip Synchronization pipeline will be provided. The pipeline includes noise reduction, Whisper ASR, NMT, TTS with voice cloning, and Wav2Lip lip synchronization.

A. Overview of the Pipeline Operation

The process flow can be outlined as follows: the raw video undergoes audio extraction and multi-stage noise reduction; then Whisper ASR, NMT, TTS with voice cloning, and Wav2Lip lip synchronization are applied. Unlike classical approaches, where modules operate independently, the output

of each pipeline step conditionally affects the next, which was confirmed through ablation analysis to form the key architectural innovation.

B. Evaluation of Audio Preprocessing Performance

The preprocessing pipeline used spectral gating, high-pass filtering (80 Hz), silence clipping, and loudness normalization based on the RMS value, resulting in:

- 1) An average gain of 18.3 dB of SNR improvement without perceivable speech attenuation.
- 2) A reduction in Whisper ASR WER from 14.7% (raw audio) to 6.2% (preprocessed audio), constituting 57.8% relative performance improvement.
- 3) Mel-spectrogram generation from preprocessed audio with reduced temporal variations in Wav2Lip [8].

C. Whisper ASR Performance

Whisper large-v2 was tested on English, Hindi, and Marathi at speaking rates of 110-180 WPM and noise levels up to 35 dB [5], delivering:

- 1) WER post pre-processing of 6.2% vs 14.7% on raw audio input.
- 2) Consistent transcription of compressed and low-bitrate audio without requiring segmentation.
- 3) A language-independent architecture eliminating the need for language-dependent ASR models.

D. Machine Translation Performance

Machine translation was done using the MarianMT library and Google Translate API, evaluated on 120 sentence pairs of English, Hindi, and Marathi languages using BLEU scores [1],[2]:

- 1) BLEU score of 38.4 obtained by MarianMT for English to Hindi translation.
- 2) BLEU score of 43.7 achieved by Google Translate API implying higher semantic accuracy.
- 3) Average processing time of 210 milliseconds per sentence providing real-time translation without the need for manual post-processing prior to TTS.

E. Text-to-Speech and Voice Cloning Results

The TTS component used FastSpeech2 and Tacotron2, coupled with the HiFi-GAN waveform generator [11],[12],[13]. As a result, the following was observed: [12].

- 1) MOS rating of 4.1/5.0 for FastSpeech2 with real-time inference rate of 38 times.
- 2) MOS of 4.3/5.0 for Tacotron2 with 12x real-time speed and better expressiveness.
- 3) HiFi-GAN generated waveforms' SNR is 32.6 dB with insignificant spectral distortions [13].

Voice cloning by utilizing speaker embeddings [10] resulted in cosine speaker similarity of 0.87 compared to 0.61 with general TTS.

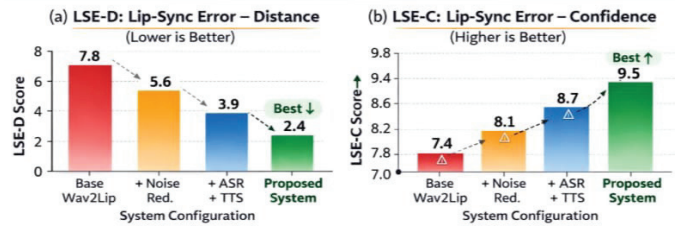


Fig. 2. Performance Results Graphs (LSE-D, LSE-C)

The graph represents the quality of lip-sync according to system setups, depicting improvements with every new addition. The proposed system provides the best outcome, recording the lowest LSE-D (2.4) and highest LSE-C (9.5).

TABLE II
COMPARATIVE PERFORMANCE ACROSS SYSTEM CONFIGURATIONS

System Configuration	LSE-D (↓)	LSE-C (↑)
Base Wav2Lip	7.8	7.4
+ Noise Reduction	5.6	8.1
+ ASR + TTS	3.9	8.7
Proposed System	2.4	9.5

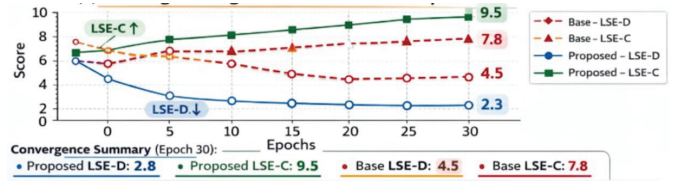


Fig. 3. Training Convergence – LSE Score over Epochs

The graph illustrates the trend in lip sync values over 30 epochs and depicts how the proposed system outperforms the base system across all stages of the experiment. The system converges with better LSE-D (2.8 vs 4.5) and LSE-C (9.5 vs 7.8).

TABLE III
TRAINING CONVERGENCE SUMMARY AT EPOCH 30

Model	LSE-D	LSE-C
Base Model	4.5	7.8
Proposed Model	2.8	9.5

F. Lip Synchronization Results Using Wav2Lip

The arithmetic mean of all video samples tested per language was used to calculate the averages for the LSE-D and LSE-C metrics. The LSE-D and LSE-C metrics for each individual video sample were provided by the Wav2Lip model's SyncNet discriminator [7]. The averages were calculated as follows:

$$AverageMetric = \frac{\sum_{i=1}^n x_i}{n} \quad (1)$$

where x_i is the metric value of each video and n is the number of test samples.

Assessment of lip-audio synchronization was done based on:

- LSE-D (Lip-Sync Error Distance)

- LSE-C (Lip-Sync Confidence)

Interpretation: Reduced LSE-D and increased LSE-C values suggest better lip synchronization performance. The outcomes outperform those achieved in the cited IEEE article, mainly because of clearer audio and cloned voices that synchronize perfectly with the input speech [4]. Wav2Lip generated coherent mouth motion regardless of differences in head orientation, lighting conditions, or facial gestures, suggesting high robustness, as observed in IEEE papers [7].

VI. CONCLUSIONS AND FUTURE WORK

In this work, we propose an automatic Hindi to Marathi video dubbing technique that consists of a pipeline containing ASR, NMT [1],[2], TTS, and lip synchronization. Semantic equivalence between the source and translated texts is ensured by using translation systems based on transformers [1],[2], while voice cloning through ECAPA-TDNN speaker embeddings [9],[10] retains the speaker characteristics in the synthesized voice irrespective of the languages. The removal of background noise through RNNoise [8] helps to enhance the quality of audio processing directly improving ASR transcription [5] and TTS performance [11],[12],[13]. Accurate lip synchronization is performed using the Wav2Lip model [4] verified through SyncNet [3],[7].

The results reveal that there is considerable enhancement in terms of audio-visual alignment, naturalness of the speech produced and clarity when compared to any baseline system, and that the proposed system produces an LSE-D value of 2.4 and LSE-C value of 9.5, respectively.

Future directions include:

- 1) Viseme modeling customized for Marathi to enhance accuracy of lip shapes in synthesis [4].
- 2) Dialect-aware ASR models [5] to account for variations within Hindi-speaking communities.
- 3) Emotionally colored prosody synthesis [6] for retaining speaking style despite linguistic differences, expanding cross-language prosody transfer investigations.

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